



BACHELOR OF INFORMATION TECHNOLOGY (HONS)

FINAL EXAMINATION FEBRUARY 2025

Course: EC3123 (Operating Systems Principles)

Time: 8.00AM – 11.00AM

(3 hours)

Lecturer: Mohd Shahdi Bin Ahmad

Date: 2 June 2025

Instructions:

Answer **ALL** questions in the Answer Booklet provided.

The maximum number of marks is 100.

This question paper consists of **2** printed pages
(excluding front cover)

DO NOT OPEN THIS BOOKLET UNTIL YOU ARE TOLD TO DO SO

1. a. Explain **THREE (3)** differences between a program and a process. (6 MARKS)
- b. Describe the **FIVE (5)** states of process. (10 MARKS)
- c. Draw and label a process creation flow diagram that shows the relationship and sequence of the system calls. (7 MARKS)
- d. What is Multithreading? (2 MARKS)
2. a. Explain the basic concepts of static and dynamic memory allocation. (4 MARKS)
- b. Illustrate the swapping process used in memory management. (8 MARKS)
- b. Illustrate the single partition and multiple partition memory allocation in contiguous memory management. (8 MARKS)
- c. Identify **FIVE (5)** segments in a logical unit. (5 MARKS)
3. a. Define the goals of I/O software. (5 MARKS)
- b. Identify the layered structure of the I/O software system. (5 MARKS)
- c. Draw a single-bus architecture in memory-mapped I/O. (6 MARKS)
- d. Draw a dual-bus memory architecture in memory mapped I/O. (7 MARKS)
- e. Identify **TWO (2)** examples of human-readable devices. (2 MARKS)
4. a. Write a Linux command based on the instructions below.

i. Create a folder named "Computing" (2 MARKS)

ii. Remove the non-empty folder named "Department". (2 MARKS)

b. Based on Table 1. answer the following questions:

No.	Group Name	Group ID	Username	User ID
1	Designer	3000	Jeff	3001
2	Programmer	4000	Lim	4001
3	Administrator	5000	Hamse	5001
4	Administrator	5000	Remy	5002
5	Finance	6000	Mary	6001
6	Finance	6000	Nani	6002

i. Show Linux commands for creating all the groups. (8 MARKS)

ii. Show Linux commands for creating all the usernames. (12 MARKS)

c. What is the command to edit a file? (1 MARK)

-END OF QUESTION PAPER-